

MinImageScorer: Diagnosing Resolution Ceilings in Agricultural Object Detection

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Abstract—We introduce a compact diagnostic, MinImageScorer (MIS), to detect whether a dataset is dominated by resolution-limited objects that yield near-zero AP in modern detectors. On Global Wheat Head Detection 2021 (GWHD), MIS identifies images enriched for very-small and small-size buckets where baseline AP is near zero, indicating a structural resolution ceiling that limits gains from pixel-duplication augmentation. We provide code and a reproducible analysis.

V. CONCLUSION

MIS is a compact, reproducible check to decide whether selection-based augmentation is likely to help. We release the notebook and the compliant figure in the repository.

I. INTRODUCTION

Small and sub-resolution objects are a common failure mode in agricultural imagery. When hard examples concentrate in regimes with near-zero baseline AP, simple augmentation strategies (e.g., copy-paste) that duplicate pixels are unlikely to increase usable information. We present a short diagnostic workflow and apply it to GWHD 2021.

II. METHOD

MIS ranks images by a lightweight difficulty score (model confidence and localization statistics) and inspects size-bucket composition using ground-truth boxes parsed from training annotations. We define near-zero AP buckets as those with $AP \leq 0.10$. We test whether the top 20% highest-score images contain a larger share of objects from near-zero AP buckets than expected by chance using a permutation test and bootstrap confidence intervals for capture share.

III. RESULTS

Baseline AP by size on GWHD 2021: $very_{small} = 0.007, small = 0.065, medium = 0.256, large = 0.383$. $Verysmall + small \approx 1.3\%$ of objects overall. The top 20% high-score images show increased concentration of near-zero objects (mean fraction 0.0157 vs 0.0079; permutation $p < 0.001$). The capture share (fraction of near-zero objects inside top 20%) is 0.2684 (expected 0.20), capture lift is 1.342x, 95% CI [0.2111, 0.3218]. Verdict: MIS correctly flags images where resolution driven failure dominates.

IV. DISCUSSION

The diagnostic shows that although tiny objects have very low AP, their small share in the dataset limits potential absolute gains from focusing only on them. When MIS indicates a resolution ceiling, next steps should prioritize methods that increase effective information (e.g., super-resolution, domain-conditioned synthesis) rather than simple pixel duplication.